

OpenStax Calculus Volume 2

Section 6.2 Exercises: Properties of Power Series

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1. If $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ and $g(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$, find the power series of $\frac{1}{2}(f(x) + g(x))$ and of $\frac{1}{2}(f(x) - g(x))$.
2. If $C(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$ and $S(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$, find the power series of $C(x) + S(x)$ and of $C(x) - S(x)$.

In the following exercises, use partial fractions to find the power series of each function.

3. $\frac{4}{(x-3)(x+1)}$
4. $\frac{3}{(x+2)(x-1)}$
5. $\frac{5}{(x^2+4)(x^2-1)}$
6. $\frac{30}{(x^2+1)(x^2-9)}$

In the following exercises, express each series as a rational function.

7. $\sum_{n=1}^{\infty} \frac{1}{x^n}$
8. $\sum_{n=1}^{\infty} \frac{1}{x^{2n}}$
9. $\sum_{n=1}^{\infty} \frac{1}{(x-3)^{2n-1}}$
10. $\sum_{n=1}^{\infty} \left(\frac{1}{(x-3)^{2n-1}} - \frac{1}{(x-2)^{2n-1}} \right)$

The following exercises explore applications of annuities.

11. Calculate the present values P of an annuity in which \$10,000 is to be paid out annually for a period of 20 years, assuming interest rates of $r = 0.03$, $r = 0.05$, and $r = 0.07$.
12. Calculate the present values P of annuities in which \$9,000 is to be paid out annually perpetually, assuming interest rates of $r = 0.03$, $r = 0.05$, and $r = 0.07$.
13. Calculate the annual payouts C to be given for 20 years on annuities having present value \$100,000, assuming respective interest rates of $r = 0.03$, $r = 0.05$, and $r = 0.07$.

14. Calculate the annual payouts C to be given perpetually on annuities having present value \$100,000, assuming respective interest rates of $r = 0.03$, $r = 0.05$, and $r = 0.07$.
15. Suppose that an annuity has a present value $P = 1$ million dollars. What interest rate r would allow for perpetual annual payouts of \$50,000?
16. Suppose that an annuity has a present value $P = 10$ million dollars. What interest rate r would allow for perpetual annual payouts of \$100,000?

In the following exercises, express the sum of each power series in terms of geometric series, and then express the sum as a rational function.

17. $x + x^2 - x^3 + x^4 + x^5 - x^6 + \cdots$ (Hint: Group powers x^{3k} , x^{3k-1} , and x^{3k-2} .)
18. $x + x^2 - x^3 - x^4 + x^5 + x^6 - x^7 - x^8 + \cdots$ (Hint: Group powers x^{4k} , x^{4k-1} , etc.)
19. $x - x^2 - x^3 + x^4 - x^5 - x^6 + x^7 - \cdots$ (Hint: Group powers x^{3k} , x^{3k-1} , and x^{3k-2} .)
20. $\frac{x}{2} + \frac{x^2}{4} - \frac{x^3}{8} + \frac{x^4}{16} + \frac{x^5}{32} - \frac{x^6}{64} + \cdots$ (Hint: Group powers $(\frac{x}{2})^{3k}$, $(\frac{x}{2})^{3k-1}$, and $(\frac{x}{2})^{3k-2}$.)

In the following exercises, find the power series of $f(x)g(x)$ given f and g as defined.

21. $f(x) = 2 \sum_{n=0}^{\infty} x^n$, $g(x) = \sum_{n=0}^{\infty} nx^n$
22. $f(x) = \sum_{n=1}^{\infty} x^n$, $g(x) = \sum_{n=1}^{\infty} \frac{1}{n} x^n$. Express the coefficients of $f(x)g(x)$ in terms of $H_n = \sum_{k=1}^n \frac{1}{k}$.
23. $f(x) = g(x) = \sum_{n=1}^{\infty} (\frac{x}{2})^n$
24. $f(x) = g(x) = \sum_{n=1}^{\infty} nx^n$

In the following exercises, differentiate the given series expansion of f term-by-term to obtain the corresponding series expansion for the derivative of f .

25. $f(x) = \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$
26. $f(x) = \frac{1}{1-x^2} = \sum_{n=0}^{\infty} x^{2n}$

In the following exercises, integrate the given series expansion of f term-by-term from zero to x to obtain the corresponding series expansion for the integral of f .

27. $f(x) = \frac{2x}{(1+x^2)^2} = \sum_{n=1}^{\infty} (-1)^{n-1} (2n) x^{2n-1}$
28. $f(x) = \frac{2x}{1+x^2} = 2 \sum_{n=0}^{\infty} (-1)^n x^{2n+1}$

In the following exercises, evaluate each infinite series by identifying it as the value of a derivative or integral of geometric series.

29. Evaluate $\sum_{n=1}^{\infty} \frac{n}{2^n}$ as $\frac{1}{2} f'(\frac{1}{2})$, where $f(x) = \sum_{n=0}^{\infty} x^n$.
30. Evaluate $\sum_{n=1}^{\infty} \frac{n}{3^n}$ as $\frac{1}{3} f'(\frac{1}{3})$, where $f(x) = \sum_{n=0}^{\infty} x^n$.
31. Evaluate $\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}$ as $\frac{1}{4} f''(\frac{1}{2})$, where $f(x) = \sum_{n=0}^{\infty} x^n$.
32. Evaluate $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ as $\int_0^1 f(t) dt$ where $f(x) = \sum_{n=0}^{\infty} (-1)^n x^{2n} = \frac{1}{1+x^2}$.

In the following exercises, given that $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, use term-by-term differentiation or integration to find power series for each function centered at the given point.

33. $f(x) = \ln x$ centered at $x = 1$ (Hint: $x = 1 - (1 - x)$.)
34. $\ln(1 - x)$ at $x = 0$
35. $\ln(1 - x^2)$ at $x = 0$
36. $f(x) = \frac{2x}{(1-x^2)^2}$ at $x = 0$
37. $f(x) = \tan^{-1}(x^2)$ at $x = 0$
38. $f(x) = \ln(1 + x^2)$ at $x = 0$
39. $f(x) = \int_0^x \ln t \, dt$ where $\ln(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-1)^n}{n}$
40. [T] Evaluate the power series expansion $\ln(1 + x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ at $x = 1$ to show that $\ln(2)$ is the sum of the alternating harmonic series. Use the alternating series test to determine how many terms of the sum are needed to estimate $\ln(2)$ accurate to within 0.001, and find such an approximation.
41. [T] Subtract the infinite series of $\ln(1 - x)$ from $\ln(1 + x)$ to get a power series for $\ln(\frac{1+x}{1-x})$. Evaluate at $x = \frac{1}{3}$. What is the smallest N such that the N th partial sum of this series approximates $\ln(2)$ with an error less than 0.001?

In the following exercises, using a substitution if indicated, express each series in terms of elementary functions and find the domain of convergence of the sum.

42. $\sum_{k=0}^{\infty} (x^k - x^{2k+1})$
43. $\sum_{k=1}^{\infty} \frac{x^{3k}}{6k}$
44. $\sum_{k=1}^{\infty} (1 + x^2)^{-k}$ using $y = \frac{1}{1+x^2}$
45. $\sum_{k=1}^{\infty} 2^{-kx}$ using $y = 2^{-x}$
46. Show that, up to powers x^3 and y^3 , $E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ satisfies $E(x + y) = E(x)E(y)$.
47. Differentiate the series $E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ term-by-term to show that $E(x)$ is equal to its derivative.
48. Show that if $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is a sum of even powers, that is, $a_n = 0$ if n is odd, then $F(x) = \int_0^x f(t) \, dt$ is a sum of odd powers, while if f is a sum of odd powers, then F is a sum of even powers.
49. [T] Suppose that the coefficients a_n of the series $\sum_{n=0}^{\infty} a_n x^n$ are defined by the recurrence relation $a_n = \frac{a_{n-1}}{n} + \frac{a_{n-2}}{n(n-1)}$. For $a_0 = 0$ and $a_1 = 1$, compute and plot the sums $S_N = \sum_{n=0}^N a_n x^n$ for $N = 2, 3, 4, 5$ on $[-1, 1]$.
50. [T] Suppose that the coefficients a_n of the series $\sum_{n=0}^{\infty} a_n x^n$ are defined by the recurrence relation $a_n = \frac{a_{n-1}}{\sqrt{n}} - \frac{a_{n-2}}{\sqrt{n(n-1)}}$. For $a_0 = 1$ and $a_1 = 0$, compute and plot the sums $S_N = \sum_{n=0}^N a_n x^n$ for $N = 2, 3, 4, 5$ on $[-1, 1]$.

51. [T] Given the power series expansion $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$, determine how many terms N of the sum evaluated at $x = -1/2$ are needed to approximate $-\ln(2)$ accurately to within $1/1000$. Evaluate the corresponding partial sum $\sum_{n=1}^N (-1)^{n-1} \frac{x^n}{n}$.
52. [T] Given the power series expansion $\tan^{-1}(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1}$, use the alternating series test to determine how many terms N of the sum evaluated at $x = 1$ are needed to approximate $\tan^{-1}(1) = \frac{\pi}{4}$ accurately to within $1/1000$. Evaluate the corresponding partial sum $\sum_{k=0}^N (-1)^k \frac{x^{2k+1}}{2k+1}$.
53. [T] Recall that $\tan^{-1}(\frac{1}{\sqrt{3}}) = \frac{\pi}{6}$. Assuming an exact value of $\frac{1}{\sqrt{3}}$, estimate $\frac{\pi}{6}$ by evaluating partial sums $S_N(\frac{1}{\sqrt{3}})$ of the power series expansion $\tan^{-1}(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1}$ at $x = \frac{1}{\sqrt{3}}$. What is the smallest number N such that $6S_N(\frac{1}{\sqrt{3}})$ approximates π accurately to within 0.001 ? How many terms are needed for accuracy to within 0.00001 ?